

GAME: Governance for Agile Management of Enterprises

A Management Model for Agile Governance

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Abstract— Agility at the business and organizational levels presents a challenge for many enterprises. Business agility demands the ability to sense and respond to changes in competitive environments, whereas organizational agility demands the dexterity to sense broader market opportunities and respond with changes that are organization-wide. These challenges require an information and communication technologies (ICT or IT) environment flexible and customizable simultaneously with the coordination across multiple organization units, also demands effective and responsive governance in order to deliver faster, better, and cheaper value to the business. Driven by these challenges, the goals of the candidate's PhD thesis is to conceive, define, and evaluate a management model for agile governance on global development environments. Preliminary insights suggest the model's format should be a corporative game, applying constructs of *gamification*, fun theory and game theory, generating a practical approach to facilitate the implementation of a collaborative and adaptive culture, in order to establish relational integration mechanisms as well as a better understanding of how these arrangements can help the organizations to attain greater enterprise agility and support its overall strategy.

Keywords— Information Systems, Agile Governance, IT management, Organizational behavior, Software Engineering.

Concentration areas: Software Engineering, Information Systems, IT Management, IT Governance

I. INTRODUCTION

Advances in information and communication technologies (ICT or IT) have allowed business to automate the design, sales and support of their products and services. In fact, IT can also be seen as a decisive component to enables enterprise agility or the ability to sense and respond to changes in the competitive business environment.

Actually, the design and maintenance of the IT for enterprise agility can be a challenge when the competitiveness of organization's products and services is depending of the application of models and frameworks that have no guidance details of how to implement and deploy the necessary management instruments and governance mechanism, such as ITIL, COBIT, among others [1], [2].

Indeed, the lack of practical approach by some models related with governance and IT management generates several questions and difficulties to the IT and business teams, such as: i) where to begin?; ii) how to adapt?; iii) what to prioritize? [3]

At the same time, these issues lead the enterprises to hire several specialized consulting services, causing, in most cases,

high investments, lengthy deployments, and generating the long-term external dependencies to the organization [4], [5].

Consequently, the challenges become even greater when dealing with these matters on a global software development and distributed environment, where cultural differences, awareness and communication style, if not treated properly can lead to conflicts. Arguably, in Global Development Environments governance issues are even more relevant and necessary, as well as its implementation even greater challenging [6].

II. RESEARCH OBJECTIVE

The objective of this candidate's PhD thesis is to address these challenges through conception, definition, and evaluation of a management model for agile governance, guided by the following research question:

How should be a management model, which can be easily assimilated by the organizational culture, and can help to reduce effort and increase the results of the process of deploying responsibilities and governance mechanisms, in a global, distributed and heterogeneous environment, in order to deliver faster, better, and cheaper value to the business?

In the subsequent sections this paper gives an overview of the related theoretical framework, the methodology adopted to elaborate and evaluate the model, preliminary results and expected results.

III. BACKGROUND AND RELATED WORK

This work is based on six areas of influence: General Administration Theory, General Systems Theory, Agile Governance, *Gamification*, Fun Theory and Game Theory.

A. Agile Governance

Corporate governance is the set of processes, customs, policies, laws and institutions affecting the way a corporation is directed, administered or controlled. Corporate governance also includes the relationships between the various parties involved and the purposes for which a society is governed [7].

On the other hand, ICT Governance is defined as a subset of Corporate Governance focused discipline on information and communication technologies and their performance systems and risk management [8].

Additionally, the Agile Methodologies [9] have been disseminated more and adding competitiveness and dynamism to the process of software development in the area of Software

Engineering, through initiatives where the principles of communication and collaboration are essential [10], [11].

Moreover, Dubinsky & Kruchten [12], [6] highlight that Software Development Governance (SDG) has emerged in the last few years to deal with establishing the structures, policies, controls, and measurements for communication and for decision rights, to ensure the success of software development organizations.

Recently, it has emerged the proposal of Agile Governance [13], [14], which provides the wide application of principles and values of Agile Software Development [15] to the conventional governance processes. As a consequence of this original work, it was proposed by De Oliveira Luna [16] the creation of a reference model for this concept, thus resulting in the agile framework: MAnGve.

However, the MAnGve is focused only in the deployment process. As a *catalyzer*, the MAnGve, accelerates the Governance deployment, but has no answers for the other aspects of the IT environment. Hence, this research considers this *gap* as an opportunity to expand and explore the paradigm of Agile Governance, by proposing a management model for global development environments.

B. General Administration Theory and General Systems Theory

According to Birkinshaw & Goddard [17], a management model is defined as a system of rules governing the business direction or set of goals delineators for a particular program of management action, and conditioners for its implementation, or even, the means by which a company seeks to adjust or modify their position in the market.

In keeping with Porter [18], management models determine the relationship between the human element and the business structure formally established to become the agent capable to providing an effective organization. Thus, whatever the management models adopted the act of structuring and organizing becomes essential for an enterprise to survive in a competitive economic environment.

Furthermore, as indicated by Drucker [19], organizational quality should occur as a whole and be a consequence of the degree of interdependence of the components and participants in the management model adopted. Therefore, the efforts of commitment to the success should be seen as a result a restructuring of work styles, and wherefore a definition of functional strategies and operationalization of corporate policies, targeted to the vision of the organization's summit.

In addition, several authors [20], [21] have suggested analyzing organizational structures through a theory of systems capable of providing a wide and comprehensive view of the decision-making, because in their view, the process of making decisions becomes easier when it is known the system within which these decisions should be taken.

C. Fun Theory, Gamification and Game Theory

Theory of fun is about how to use ludic attributes to motivate people to do edifying (good) things, probably increasing productivity and generating satisfaction and enjoyment during activities. It is based on the concept of maximizing intrinsic reward for the active creation or discovery of novel, surprising patterns allowing for improved

prediction or data compression. It generalizes the traditional field of active learning, and is related to old, but less formal ideas in aesthetics theory and developmental psychology [22].

According to Deterding *et al.* [23] the most recent phenomenon in Information Systems trajectory is '*gamification*', an umbrella term for the use of game elements (rather than full-fledged games) to improve user experience and user engagement in non-game services and applications. In other words, *gamification* is about how to use game thinking and game mechanisms in a non-game context, in order to engage people and solve problems.

Game Theory aims to help understand situations in which decision-makers interact, something closely related with Governance. However, its scope is vastly larger, and devotes very little space to games in the everyday sense; its main focus is the use of game theory to illuminate economic, social, political, and biological phenomena [24].

Eventually, the combination of these approaches can be applied as an instrument in order to reduce the risks and challenges related with the governance issues in a global software development and distributed environment.

IV. RESEARCH METHOD AND PRELIMINARY RESULTS

The definition of the methodological tools is directly related to the problem being studied. The methodological framework of reference, when carefully selected, is what gives scientific rigor to a research [25].

In fact, in this work, a combination of methods and techniques will be applied in several crucial moments of its development, which can include: review of the state of the art of agile governance, review about management models in the context of the General Administration Theory and General Systems Theory; review of the theoretical constructs that can be used on the model assembly in the context of: *Gamification*, Fun Theory and Game Theory; among others. At last, in summary the methodological framework of this research is summarized below in Table I.

TABLE I - CLASSIFICATION OF METHODOLOGICAL APPROACH [25].

About the objective	Exploratory
About the technical procedure	Literature Review, Systematic Mapping Review, Bibliometrics and Scientometrics
About the nature of variables	Qualitative
About method approach	Inductive
About methods of procedure	Comparative and Structuralist.

A. Model Development

In addition, according to the objective of this research it can be classified as exploratory, as it aims to present the proposal for a management model for Agile Governance. Furthermore, the preparation of the proposed model will be based on an inductive approach, supported by structuralist and comparative methods. Meanwhile, the use of such methods of procedure will be essential to conduct a qualitative analysis of information obtained in a literature review [26].

B. Model Evaluation

This work intends to use the procedure of case study to evaluate the application of the model in a real organization. A case study is an intensive analysis of an individual unit (e.g., a person, group, enterprise, or event) stressing developmental

factors in relation to context. This research intends to apply instrumental case studies for evaluate the proposed model. This type of case study is used to provide insight into an issue or helps to refine a theory or model [27].

C. Preliminary Results

As one of the initiatives of this research, is being conducted a Systematic Mapping Literature Review about the state of art of Agile Governance in the world. This study should return a set of theoretical constructs to contribute with the model core definition, such as: concepts, principles, practices, problems, challenges, benefits, motivations, among others. The preliminary results about this study are promising, and should generate a relevant publication soon. Indeed, these evidences, lead us to believe that *agile governance* can be *widely* and *holistically* defined, as:

“the ‘means’ by which strategic competitive advantages ought to be achieved and improved on the organizational environment, under an agile approach in order to deliver faster, better, and cheaper value to the business.”

At the same time, as a complementary study line is being conducted an Exploratory Literature Review about Management Models in the context of General Administration Theory and General Systems Theory. This another study should contribute with a set of theoretical constructs to the design of the model, such as: approach, usefulness, applicability, shortcomings, limitations, critical success factors, among others.

Furthermore, is being started another Exploratory Literature Review about how *Gamification*, Fun Theory and Game Theory, can be applied to the model assembly in order to overcome the organizational barriers and potentiate the natural assimilation and internalization of the model being proposed.

V. EXPECTED RESULTS

The authors believe that not only software development organizations, but also whole IT industry will benefit with the results of this research. In fact, improving the competitiveness of governments and companies through the improvement of their governance and management shall result in significant economic returns. In a broader context, they believe that it will help any kind of organization to deliver faster, better, and cheaper value to the business.

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REFERENCES

- [1] L. Gerke and G. Ridley, “Tailoring CobiT for Public Sector IT Audit: An Australian Case Study,” in *Information technology governance and service management: frameworks and adaptations*, 1st ed., K. Klinger, Ed. Hershey: Information Science Reference (an imprint of IGI Global), 2009, pp. 101–125.
- [2] T. Mendel, “ITIL’s Final Breakthrough: From ‘What’ to ‘How’,” *CSO Online*, pp. 1–3, 2004.
- [3] G. Spafford and G. Kim, “Top ITIL Myths,” *Internet News.com*, pp. 2–3, 2004.
- [4] ISACA, “COBIT Case Studies by Industry,” 2013. [Online]. Available: <http://www.isaca.org/Knowledge-Center/cobit/Pages/COBIT-Case-Studies.aspx>. [Accessed: 27-May-2013].
- [5] M. Fry, “Top Ten Reasons Organizations are Unsuccessful Implementing ITIL,” *TechRepublic*, pp. 1–8, 2005.
- [6] Y. Dubinsky, S. Ravid, A. Rafaeli, and R. Bar-Nahor, “Governance Mechanisms in Global Development Environments,” *2011 IEEE Sixth International Conference on Global Software Engineering*, pp. 6–14, Aug. 2011.
- [7] A. Cadbury, “The Financial Aspects of Corporate Governance,” London, 1992.
- [8] IT Governance Institute, *Board briefing on IT governance*, 2nd ed. Rolling Meadows: IT Governance Institute, 2001, p. 66.
- [9] T. Dybå and T. Dingsøyr, “Empirical studies of agile software development: A systematic review,” *Information and Software Technology*, vol. 50, no. 9–10, pp. 833–859, Aug. 2008.
- [10] P. Kruchten, “Contextualizing agile software development,” *Journal of Software Maintenance and Evolution*, pp. 1–12, 2011.
- [11] Y. Dubinsky, A. Yaeli, and E. Zarpas, “Governance of software development: The transition to agile scenario,” in *Software Applications: Concepts, Methodologies, Tools, and Applications*, P. Tiako, Ed. Hershey, PA: Information Science Reference, 2008, pp. 309–327.
- [12] Y. Dubinsky and P. Kruchten, “(POP-038) 2nd workshop on software development governance (SDG),” *2009 31st International Conference on Software Engineering - Companion Volume*, pp. 455–456, 2009.
- [13] A. J. H. De Oliveira Luna, C. P. Costa, H. P. De Moura, and M. A. Novaes, “(POP-009) Agile Governance in Information and Communication Technologies: Shifting Paradigms,” *JISTEM Journal of Information Systems and Technology Management*, vol. 7, no. 2, pp. 311–334, 2010.
- [14] F. A. Cummins, *Building the Agile Enterprise: With SOA, BPM and MBM*, 1st ed. Burlington: Elsevier Science, 2009, p. 336.
- [15] K. Beck, M. Beedle, A. van Bennekum, A. Cockburn, W. Cunningham, M. Fowler, J. Grenning, J. Highsmith, A. Hunt, R. Jeffries, J. Kern, B. Marick, R. C. Martin, S. Mellor, K. Schwaber, J. Sutherland, and D. Thomas, “Manifesto for Agile Software Development,” 2001. [Online]. Available: <http://agilemanifesto.org/>. [Accessed: 01-May-2013].
- [16] A. J. H. De Oliveira Luna, “MANGve: A model for Agile Governance in ICT,” *Federal University of Pernambuco*, 2009.
- [17] J. Birkinshaw and J. Goddard, “What is your management model?,” *MIT Sloan Management Review*, no. 50216, 2009.
- [18] M. E. Porter, *Competitive Strategy: Techniques for Analyzing Industries and Competitors*. Free Press, 1980, p. 432.
- [19] P. F. Drucker, *The frontiers of management*, First Edit. New York: Truman Talley Books, 1986, p. 553.
- [20] W. Hofkirchner, “Ludwig von Bertalanffy. Forerunner of evolutionary systems theory,” *IFSR 2005 : Proceedings of the First World Congress of the International Federation for Systems Research*, p. 6, 2005.
- [21] L. Von Bertalanffy, “The History and Status of General Systems Theory,” *Academy of Management Journal*, vol. 15, no. 4, pp. 407–426, 1972.
- [22] J. Schmidhuber, “Formal theory of creativity, fun, and intrinsic motivation (1990–2010),” *Autonomous Mental Development, IEEE Transactions*, vol. 2, no. 3, pp. 230–247, 2010.
- [23] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, “From game design elements to gamefulness: defining gamification,” *Proceedings of the 15th International Academic MindTrek Conference Envisioning Future Media Environments*, 2011.
- [24] C. F. Camerer, “Progress in Behavioral Game Theory,” *Journal of Economic Perspectives*, vol. 11, no. 4, pp. 167–188, Nov. 1997.
- [25] B. Gower, *Scientific method: A historical and philosophical introduction*. London and New York: Taylor & Francis e-Library, 2002, p. 288.
- [26] I. Jebreen, “Using Inductive Approach as Research Strategy in Requirements Engineering,” *International Journal of Computer and Information Technology*, vol. 01, no. 02, pp. 162–173, 2012.
- [27] R. K. Yin, *Case Study Research: Design and Methods*. SAGE Publications, 2003, p. 181.